Lab Class ML:I, ML:II

Until Wednesday, Nov. 11th, 2026, 11:59 pm CET, solutions to the following exercises may be submitted as one zip-file named `ml26-ex1-group<your-group-number>.zip` via Moodle.

Exercise 1 : Machine Learning (general)

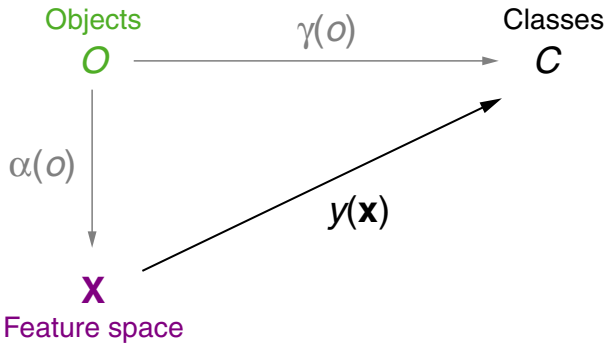
(a) Define the terms “supervised learning”, “unsupervised learning”, and “reinforcement learning”.

(b) Determine the learning paradigm (supervised, unsupervised, reinforcement) for the following tasks.
Note: If more than one learning paradigm is possible, select one and provide a brief (1 sentence) explanation.

(b1) Grouping YouTube accounts by watch behaviour
(b2) Identifying plants based on pictures
(b3) A program learning to win at chess
(b4) A robot learning to walk by feedback
(b5) Facial recognition
(b6) Discovering topics in a large corpus of research papers

Exercise 2 : Specification of Learning Tasks

The following picture from the lecture slides describes the relationship between Real World and Model World, when it comes to the specification of learning tasks.



Assume you are building a machine learning system that predicts whether a wine sample is of high or low quality. For the following list, decide which symbol from the picture most closely matches the given list item:

(a) A cellar containing hundreds of wine bottles of different types.

(b) A table listing for each wine its measured values of acidity, sugar, alcohol, and pH.

(c) A professional sommelier who can accurately rate any wine as “high-quality” or “low-quality.”

(d) A laboratory instrument that measures chemical features like acidity, sugar, and alcohol content.

(e) The set $\{\mathsf{HighQuality}, \mathsf{LowQuality}\}$

(f) The machine learning model you are trying to train that predicts wine quality from its measured features.

© Stein/Bevendorff/Kanadan 2026

Exercise 3 : [P] Data Annotation and Feature Engineering

Throughout the programming labs, we will work on the task of text regression: given a text, predict whether it was written by a human or an AI. The dataset comprises multiple text genres, such as news articles, Wikipedia intro texts, or fanfiction.

The main purpose of this exercise is to get familiar with the dataset that we will be using throughout the labs. Each group member has to read 5 texts and label each of them as written by a human or a large language model (LLM).

(a) Annotate your part of the dataset. To get the annotation data, use the "[annotation]" button on Moodle schedule. Click on "Guidelines" and read them carefully before annotating.

After annotating, click "save" to download your annotations. Add the downloaded json files to the zip file that you submit to Moodle.

Note: To get points for this exercise, all group members have to submit their annotations.

(b) Look back to the exercise 2. Which symbol in the picture corresponds to the role that **you** are playing? Which symbol corresponds to the functions that you implement in this exercise?

(c) Implement a program to extract the features from texts, such as the number of sentences and the average word length. Come up with 3 additional features that you think might be useful for this task. The program should take a text as an input and output a CSV file with columns corresponding to the five features and rows corresponding to texts.

Exercise 4 : Rule-Based Learning

The examples of a training set for a classification problem are described by the values of the attributes $A_1, \ldots, A_p$ and the related concepts $C = \{0, 1\}$. For the attributes $A_1, \ldots, A_p$ there are in each case $m_1, \ldots, m_p$ values, e.g. $a_{i,1}, \ldots, a_{i,m_i}$ for A_i. The hypothesis space contains the conjunctions of restrictions for the attributes: "A_1 has value a_{1,j_1} and ... and A_p has value a_{p,j_p}". A question mark in a hypothesis denotes a wildcard for the respective attribute domain. The hypothesis space does also contain the empty hypothesis $\perp$, which assigns all examples to the concept 0.

(a) Determine the number $n(p)$ of all possible examples for this problem.

(b) Determine the number $|H_p|$ of different hypotheses.

(c) How will the above answers change, if an additional attribute A_{p+1} with m_{p+1} values is added? Derive a recursion formula.

© Stein/Bevendorff/Kanadan 2026

Exercise 5 : Rule-Based Learning (Practice)

Given is the following training set D, which you have obtained as co-driver by observing your friend:

	Weekday	Mother-in-the-car	Mood	Time of day	run-a-red-light
1	Monday	no	easygoing	evening	yes
2	Monday	no	annoyed	evening	no
3	Saturday	yes	easygoing	lunchtime	no
4	Monday	no	easygoing	morning	yes

Let the set H contain hypotheses that are built from a conjunction of restrictions for attribute-value combinations; e. g. $(\textit{Monday}, \textit{yes}, ?, ?)$.

(a) Apply the Find-S algorithm for the example sequence 1, 2, 3, 4.

(b) Apply the Candidate-Elimination algorithm for the example sequence 1, 2, 3, 4, and identify the boundary sets H_S and H_G.

(c) What is the version space H_D for this example?

Exercise 6 : Rule-Based Learning in 2D

Consider the problem of rule-based learning in the following, rather different, feature space: The set of possible examples is given by all points of the x-y plane with integer coordinates from the interval $[1, 10]$. The hypothesis space is given by the set of all rectangles. A rectangle is defined by the points (x_1, y_1) and (x_2, y_2) (bottom left and upper right corner). Hypotheses are written as $\theta = (\textit{x}_1, \textit{y}_1, \textit{x}_2, \textit{y}_2)$, and assign a point (x, y) to the value 1, if $x_1 \leq x \leq x_2$ and $y_1 \leq y \leq y_2$ hold, with arbitrary, but fixed integer values for x_1, y_1, x_2, y_2 from the interval $[1, 10]$.

Hint: The maximally specific hypothesis h_{s_0} corresponds to a "zero-sized" rectangle that doesn't contain any points with integer coordinates; you may use the symbol $(\perp, \perp, \perp, \perp)$.

(a) For the setting described above, formulate the most general hypothesis h_{g_0}.

(b) Clarify for yourself how the "more-general" relation $\geq_g$ works in this setting, and check all that apply:

- ☐ $(\textit{1}, \textit{2}, \textit{3}, \textit{4}) \geq_g (\textit{1}, \textit{1}, \textit{4}, \textit{4})$
- ☐ $(\textit{2}, \textit{3}, \textit{6}, \textit{7}) \geq_g (\textit{3}, \textit{4}, \textit{5}, \textit{7})$
- ☐ $(\textit{1}, \textit{1}, \textit{2}, \textit{8}) \geq_g (\textit{1}, \textit{1}, \textit{3}, \textit{3})$
- ☐ $(\textit{3}, \textit{3}, \textit{9}, \textit{9}) \geq_g (\textit{1}, \textit{1}, \textit{1}, \textit{1})$

(c) Given a hypothesis $h : \theta = (\textit{2}, \textit{3}, \textit{5}, \textit{7})$, and an example $\mathbf{x} = (2, 7)$ with $c = 0$, determine two hypotheses h_1 and h_2 such that both are minimal specializations of h, and both are consistent with $(\mathbf{x}, c)$.

Hint: for the correct answers h_i, there must not exist any other hypothesis h' consistent with $(\mathbf{x}, c)$ where $h \geq_g h'$ and $h' \geq_g h_i$.

© Stein/Bevendorff/Kanadan 2026